

IVACS V-TRACE: Vehicle Trust, Route & Evidence Engine

Problem Statement: OD-08 — License Plate Detection & Recognition (Construction-Site CCTV)

Report Updated: 2026-09-18 10:37:57

Phase: Foundation Verified (Ready for Identity Engine)

Policy: Zero LLM / CV-Native

1. System Foundation Verification Status

Component / Subsystem	Verification Status
Backend (FastAPI REST Server)	PASS
MP4 Ingestion (Sequential & Configurable Sampling)	PASS
Vehicle Detection (Ultralytics YOLOv8 Nano)	PASS
Plate Detection (Bumper ROI & Vehicle Association)	PASS
OCR Engine (EasyOCR Dual Text & Normalization)	PASS
Evidence Image Storage (Frame, Vehicle, Plate Crops)	PASS
Database Engine (SQLite Persistence)	PASS
REST API Endpoints (/health, /detections, /demo/*)	PASS
Automated Tests (100% Pass Rate across 7 core suites)	PASS

2. Team Member Role Allocation (CodeAvengers Team)

Member	GitHub / Email	Assigned Subsystem Role
Jone Kavish (Lead)	jonekavish-dot jonekavish@gmail.com	Backend Architecture, Overall Team Management, EasyOCR Pipeline, API Endpoints, Frame Orchestrator
K.V. Pranesh (Member-1)	kvpranesh kvpranesh49@gmail.com	Computer Vision & Vehicle Detection Engine (YOLOv8n Inference, Multi-Class COCO Vehicle Filtering)
Gowshik Gunal (Member-2)	gowshikgunal22 gowshikgunal@gmail.com	License Plate Localization Engine (Bumper ROI Analysis, Vehicle Association, Image Preprocessing)
Dinesh Balu (Member-3)	dineshbalu7f-glitch dineshbalu7.f@gmail.com	Video Source Abstraction (MP4/RTSP/Webcam Streams), Frame Sampling, Evidence Storage & SQLite Persistence

3. Milestones Executed & Verified

Step	Milestone	Details / Output	State
Step 1	Codebase Inspection	Assessed workspace, audited Python 3.14 environment, created ARTIFACTS/current_state.md	Completed
Step 2	Dependencies Provisioning	Installed and verified opencv-python, ultralytics 8.4, torch 2.14, easyocr 1.7	Completed
Step 3	Modular Architecture	Engineered backend/ (config, video, detection, ocr, services, database, schemas)	Completed
Step 4	Video & Pipeline Integration	MP4 sequential reader, sampling control, vehicle-to-plate association, evidence storage	Completed
Step 5	Automated Testing	Executed 7 comprehensive pytest test suites covering all edge cases (100% pass)	Completed
Step 6	Live Demo & API Verification	Ran FastAPI test client with demo runner; verified /health, /demo, /detections, SQLite records	Completed
Step 7	Team-wise Git Distribution	Structured git commits per team member and prepared GitHub push	Completed